MATHEMATICAL FOUNDATIONS OF POLITICAL METHODOLOGY: PLSC 43401

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Course description

This is a first course on the theory and practice of mathematical methods in social science research. These mathematical and computer skills are needed for the quantitative and formal modeling courses offered in the political science department, and are increasingly necessary for courses in American Politics, Comparative Politics, and International Relations. We will cover mathematical techniques (linear algebra, calculus, probability) and methods of logical and statistical inference (proofs and statistics). A weekly computing lab will apply these methods, as well as introduce the R statistical computing environment.

Grading: You will be graded on weekly problem sets (worth 20% of your total grade), weekly quizzes covering the reading (also 20%), and two, noncumulative exams (each worth 30% of your grade).

Problem sets are due each Tuesday at midnight, starting October 9$^{\text{th}}$. Upload your problem sets to Canvas.

Required books: *Mathematics for Economists, 4th Edition*, by Pemberton & Rau, and *Introduction to Probability, 2nd Edition*, by Bertsekas & Tsitsiklis.

R resources: You can read about regular expressions in *Mastering Regular Expressions* by Friedl. We will make available *Foundations of R* by Gulotty, a series of R exercises for the course that are available on Canvas. Students should also work through the following courses from DataCamp.com:

- Introduction to R
- Working with the RStudio IDE
- Data Manipulation in R with dplyr
- Reporting with R Markdown
- Importing Data Into R

OCT 2: ALGEBRA, SETS, AND FUNCTIONS

- Pemberton & Rau Ch, 1-4
- Foundations of R: Chapter 1, data frames, Boolean logic, indexing, loops, functions.

OCT 4: MATHEMATICAL AND SOCIAL SCIENTIFIC REASONING

- *Propositions* and *Patterns of Proof*

OCT 9: SEQUENCES, SERIES AND LIMITS

- Pemberton & Rau Ch. 5
- Foundations of R: Chapter 2, Modeling

OCT 11: DIFFERENTIATION

- Pemberton & Rau Ch. 6, 7

OCT 16: OPTIMIZATION

- Pemberton & Rau Ch. 8, 10
- Foundations of R: Chapter 3, Plotting with ggplot2

OCT 18: INTEGRATION

- Pemberton & Rau Ch. 19, 20

OCT 23: LINEAR ALGEBRA

- Pemberton & Rau Ch. 11, 12, 13.1
- Foundations of R: Chapter 4, The data science workflow

OCT 25: MULTIVARIATE CALCULUS

- Pemberton & Rau Ch. 13, 14, 16
- Foundations of R: Chapter 5, Tidy Data

OCT 30: IN-CLASS EXAM

NOV 1: LAWS AND PROPERTIES OF PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.1-1.2

NOV 6: CONDITIONAL PROBABILITY

- Bertsekas-Tsitsiklis Ch. 1.3-1.7
- Foundations of R: Chapter 6, Vectorized code over lists

NOV 8: DISCRETE RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 2

NOV 13: CONTINUOUS RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.1-3.3
- Foundations of R: Chapter 7, Baseball!

NOV 15: SUMMARIZING RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 3.4-3.7

NOV 20: COVARIANCE AND CORRELATION

- Bertsekas-Tsitsiklis Ch. 4.1-4.3
- Foundations of R: Chapter 8, String Manipulation with Regex

NOV 27: TRANSFORMATIONS OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 4.4-4.4.6

NOV 29: INEQUALITIES AND LIMITS OF SEQUENCES OF RANDOM VARIABLES

- Bertsekas-Tsitsiklis Ch. 5
- Foundations of R: Chapter 9, Geospatial Data.

DEC 4: HYPOTHESIS TESTING

- Bertsekas-Tsitsiklis Ch. 9
- Degroot and Schervish Chapter 8
- Foundations of R: Chapter 10, Classification

DEC 6: IN-CLASS EXAM